

MASTERING DECIMALS

EXERCISE 3B COMPLETE SOLUTIONS GUIDE

SECTION 1: ADDITION OF DECIMALS

TEXTBOOK METHODOLOGY

Rule for Addition: Convert the given decimals into *like decimals* (having the same number of decimal places). Write the addends vertically under each other, making sure that their decimal points align in the same column. Add as with whole numbers, and place the decimal point in the sum directly below the points in the addends.

1 (i). Add 9.67 and 15.98

The numbers **9.67** and **15.98** already have the same number of decimal places (two decimal places). They are like decimals. Write them column-wise and add.

$$\begin{array}{r} 9.67 \\ + 15.98 \\ \hline 25.65 \end{array}$$

1 (ii). Add 6.9 and 23.84

Convert **6.9** and **23.84** to like decimals by annexing a zero to the right of 6.9: **6.90** and **23.84**. Align vertically and add.

$$\begin{array}{r} 6.90 \\ + 23.84 \\ \hline 30.74 \end{array}$$

1 (iii). Add 35.67, 18.794, 4.9, 31.82

Convert the given decimals **35.67**, **18.794**, **4.9**, **31.82** to like decimals by annexing zeros to make 3 decimal places: **35.670**, **18.794**, **4.900**, **31.820**. Align vertically and add.

$$\begin{array}{r} 35.670 \\ 18.794 \\ 4.900 \\ + 31.820 \\ \hline 91.184 \end{array}$$

1 (iv). Add 63.05, 24.839, 3.7, 16.85

Convert the decimals **63.05**, **24.839**, **3.7**, **16.85** to like decimals (3 decimal places): **63.050**, **24.839**, **3.700**, **16.850**. Align column-wise and add.

$$\begin{array}{r} 63.050 \\ 24.839 \\ 3.700 \\ + 16.850 \\ \hline 108.439 \end{array}$$

SECTION 2: SUBTRACTION OF DECIMALS

TEXTBOOK METHODOLOGY

Rule for Subtraction: Convert the given decimals into *like decimals*. Write the smaller number under the larger one in a column, aligning their decimal points. Subtract column-wise as with whole numbers, placing the decimal point in the difference directly below the points in the numbers.

2 (i). Subtract 2.975 from 8.23

Subtract **2.975** from **8.23**. First, convert 8.23 into the like decimal **8.230**. Align vertically under 8.230 and subtract.

$$\begin{array}{r} 8.230 \\ - 2.975 \\ \hline 5.255 \end{array}$$

2 (ii). Subtract 0.75 from 1.3

Subtract **0.75 from 1.3**. Convert 1.3 to the like decimal **1.30**. Align vertically and subtract.

$$\begin{array}{r} 1.30 \\ - 0.75 \\ \hline 0.55 \end{array}$$

2 (iii). Subtract 6.054 from 11.26

Subtract **6.054 from 11.26**. Convert 11.26 to the like decimal **11.260**. Align column-wise and subtract.

$$\begin{array}{r} 11.260 \\ - 6.054 \\ \hline 5.206 \end{array}$$

2 (iv). Subtract 134.68 from 201.3

Subtract **134.68 from 201.3**. Convert 201.3 to the like decimal **201.30**. Align column-wise and subtract.

$$\begin{array}{r} 201.30 \\ - 134.68 \\ \hline 66.62 \end{array}$$

3. Take out 6.345 from 8.1

To 'take out' **6.345 from 8.1** means to subtract 6.345 from 8.1. Convert 8.1 to the like decimal **8.100**. Align vertically and subtract.

$$\begin{array}{r} 8.100 \\ - 6.345 \\ \hline 1.755 \end{array}$$

4. Difference between 68.5 and 0.685

To find the difference between **68.5 and 0.685**, subtract the smaller number from the larger one. Convert 68.5 to the like decimal **68.500**, align, and subtract.

$$\begin{array}{r} 68.500 \\ - 0.685 \\ \hline 67.815 \end{array}$$

5. Excess of 90 over 53.865

The excess of **90 over 53.865** represents the difference: $90 - 53.865$. Convert 90 to the like decimal **90.000**. Align column-wise and subtract.

$$\begin{array}{r} 90.000 \\ - 53.865 \\ \hline 36.135 \end{array}$$

6. What should be subtracted from 50 to get 34.57?

Let the required number be x . Then:

$$50 - x = 34.57 \implies x = 50 - 34.57.$$

Convert 50 to the like decimal **50.00**. Align vertically and subtract.

$$\begin{array}{r} 50.00 \\ - 34.57 \\ \hline 15.43 \end{array}$$

7. What should be added to 63.47 to get 91?

Let the required number be x . Then:

$$63.47 + x = 91 \implies x = 91 - 63.47.$$

Convert 91 to the like decimal **91.00**. Align column-wise and subtract.

$$\begin{array}{r} 91.00 \\ - 63.47 \\ \hline 27.53 \end{array}$$

SECTION 3: APPLIED WORD PROBLEMS

8. Manish buys metro ticket

Total money handed over by Manish = Rs. 20.00

Cost of the metro ticket = Rs. 8.75

Money received back = Total Money - Ticket Cost
= Rs. (20.00 - 8.75).

Therefore, Manish gets back **Rs. 11.25**.

$$\begin{array}{r} 20.00 \\ - 8.75 \\ \hline 11.25 \end{array}$$

9. Weight gain of a baby

Initial weight of the baby = 5.58 kg

Increased weight of the baby = 6.16 kg

Weight gained = Final Weight - Initial Weight

= 6.16 kg - 5.58 kg.

Therefore, the weight gained is **0.58 kg**.

$$\begin{array}{r} 6.16 \\ - 5.58 \\ \hline 0.58 \end{array}$$

SECTION 4: SIMPLIFICATION OF EXPRESSIONS

TEXTBOOK METHODOLOGY

Rule for Simplification: Combine operations using standard algebraic grouping. First, convert all decimals to like decimals. Group positive terms together and negative terms together. Find the sum of positive terms and the sum of negative terms separately, then subtract the negative sum from the positive sum.

10 (i). Simplify: $67 + 13.85 - 29.904$

Simplify $67 + 13.85 - 29.904$.

Convert to like decimals: $67.000 + 13.850 - 29.904$.

Step 1 (Add positive terms): $67.000 + 13.850 = 80.850$.

Step 2 (Subtract): $80.850 - 29.904 = 50.946$.

$$\begin{array}{r} 67.000 \\ + 13.850 \\ \hline 80.850 \\ - 29.904 \\ \hline 50.946 \end{array}$$

10 (ii). Simplify: $16.753 + 4.06 - 13.89$

Simplify $16.753 + 4.06 - 13.89$.

Convert to like decimals: $16.753 + 4.060 - 13.890$.

Step 1 (Add positive terms): $16.753 + 4.060 = 20.813$.

Step 2 (Subtract): $20.813 - 13.890 = 6.923$.

$$\begin{array}{r} 16.753 \\ + 4.060 \\ \hline 20.813 \\ - 13.890 \\ \hline 6.923 \end{array}$$

10 (iii). Simplify: $1 - 10.5 + 12.213$

Simplify $1 - 10.5 + 12.213$.

Convert to like decimals: $1.000 - 10.500 + 12.213$.

Group positive terms first: $(1.000 + 12.213) - 10.500$.

Step 1 (Add positives): $1.000 + 12.213 = 13.213$.

Step 2 (Subtract negative): $13.213 - 10.500 = 2.713$.

$$\begin{array}{r} 1.000 \\ + 12.213 \\ \hline 13.213 \\ - 10.500 \\ \hline 2.713 \end{array}$$

10 (iv). Simplify: $81 - 15.68 - 4.2$

Simplify $81 - 15.68 - 4.2$.

Convert to like decimals: $81.00 - 15.68 - 4.20$.

Group negative terms first: $81.00 - (15.68 + 4.20)$.

Step 1 (Add negatives): $15.68 + 4.20 = 19.88$.

Step 2 (Subtract sum): $81.00 - 19.88 = 61.12$.

$$\begin{array}{r} 81.00 \\ - 15.68 \\ - 4.20 \\ \hline 61.12 \end{array}$$

10 (v). Simplify: 555 - 55.5 - 5.555

Simplify 555 - 55.5 - 5.555.

Convert to like decimals: 555.000 - 55.500 - 5.555.

Group negative terms: 555.000 - (55.500 + 5.555).

Step 1 (Add negatives): 55.500 + 5.555 = 61.055.

Step 2 (Subtract sum): 555.000 - 61.055 = 493.945.

$$\begin{array}{r}
 55.500 \\
 + \quad 5.555 \\
 \hline
 61.055 \\
 555.000 \\
 - \quad 61.055 \\
 \hline
 493.945
 \end{array}$$

10 (vi). Simplify: 100 - 32.5 - 46.74 - 12.925

Simplify 100 - 32.5 - 46.74 - 12.925.

Convert to like decimals: 100.000 - 32.500 - 46.740 - 12.925.

Group all negative terms: 100.000 - (32.500 + 46.740 + 12.925).

Step 1 (Add negatives): 32.500 + 46.740 + 12.925 = 92.165.

Step 2 (Subtract sum): 100.000 - 92.165 = 7.835.

$$\begin{array}{r}
 32.500 \\
 46.740 \\
 + \quad 12.925 \\
 \hline
 92.165 \\
 100.000 \\
 - \quad 92.165 \\
 \hline
 7.835
 \end{array}$$